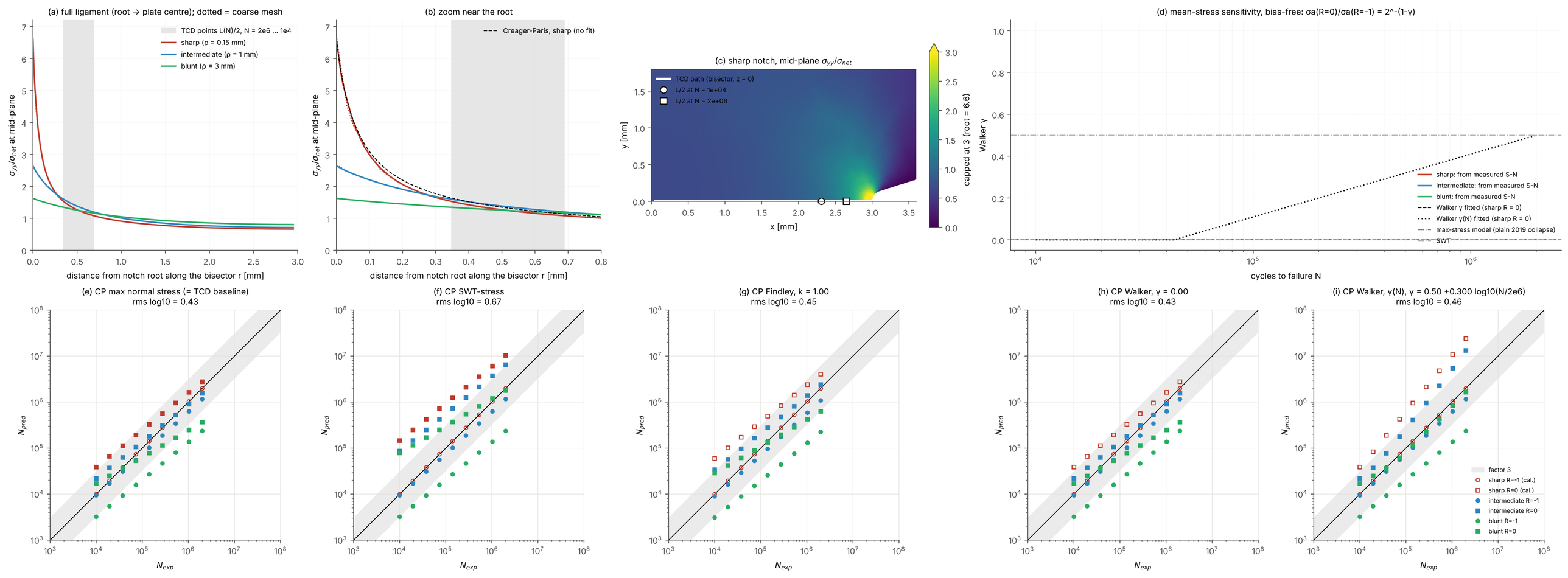

(a) full ligament (root → plate centre); dotted = coarse mesh
TCD points L(N)/2, N = 2e6 ... 1e4
sharp (ρ = 0.15 mm)
intermediate (ρ = 1 mm)
blunt (ρ = 3 mm)
σyy/σnet at mid-plane
distance from notch root along the bisector r [mm]
(b) zoom near the root
Creager-Paris, sharp (no fit)
σyy/σnet at mid-plane
distance from notch root along the bisector r [mm]
(c) sharp notch, mid-plane σyy/σnet
TCD path (bisector, z = 0)
L/2 at N = 1e+04
L/2 at N = 2e+06
y [mm]
x [mm]
capped at 3 (root = 6.6)
(d) mean-stress sensitivity, bias-free: σa(R=0)/σa(R=-1) = 2^-(1-γ)
Walker γ
cycles to failure N
sharp: from measured S-N
intermediate: from measured S-N
blunt: from measured S-N
Walker γ fitted (sharp R = 0)
Walker γ(N) fitted (sharp R = 0)
max-stress model (plain 2019 collapse)
SWT
(e) CP max normal stress (= TCD baseline)
rms log10 = 0.43
(f) CP SWT-stress
rms log10 = 0.67
(g) CP Findley, k = 1.00
rms log10 = 0.45
(h) CP Walker, γ = 0.00
rms log10 = 0.43
(i) CP Walker, γ(N), γ = 0.50 +0.300 log10(N/2e6)
rms log10 = 0.46
Npred
Nexp
factor 3
sharp R=-1 (cal.)
sharp R=0 (cal.)
intermediate R=-1
intermediate R=0
blunt R=-1
blunt R=0